

How server-on-Edge AI works?

The server-on-edge AI architecture operates through a series of interconnected processes. It begins with video stream input, where security cameras capture real-time video feeds and transmit them to a central server for processing. Once the footage reaches the server, it undergoes AI processing, which involves three key stages: preprocessing (including filtering, noise reduction, and frame extraction), inference (where AI models analyse the video to detect potential threats or anomalies), and postprocessing (which generates actionable insights such as alerts or detailed reports). Following analysis, the storage and retrieval phase ensures that processed data is securely stored in structured formats like SQL databases or object storage systems, facilitating easy access for historical analysis. Finally, in the event that a threat is identified, the system initiates the alerting and response mechanism, automatically notifying security personnel through applications, emails, or alarm systems to ensure swift action.

advances in surveillance capabilities. These cameras can recognise patterns, detect anomalies, and identify specific objects or individuals—thereby significantly strengthening security systems. The Hanwha Techwin Wisenet P Series offers AI-powered surveillance with deep learning-based analytics. These cameras detect and classify people, vehicles, and faces in real time, while minimising false alarms. With up to 4K resolution, wide dynamic range, and excellent low-light performance, they ensure clarity in all conditions. AI metadata tagging further enhances search efficiency and supports faster, smarter investigations.

One major breakthrough is on-device learning—or edge AI—where cameras process data locally instead of relying on the cloud. This shift reduces latency and enhances data privacy. With edge processors like NVIDIA Jetson or Google Coral, AI-enabled cameras can:

- Adapt to new numbers or formats (like a local license plate change).
- Train on unique environments, understanding what's 'normal' and flagging anomalies.
- Refine accuracy over time, reducing false alarms and missed detections.

AI cameras are fast evolving and offer features far beyond traditional surveillance. Highlights include:

Smart analytics. AI cameras analyse video feeds in real time to assess crowd density, detect

abnormal behaviour, or even conduct sentiment analysis. Businesses gain actionable insights—for example, retailers can optimise layouts based on customer movement patterns.

Smarter focus. These cameras learn the layout of a space, adjusting their field of view automatically to improve coverage in dynamic environments—whether in a city street or a remote location.

Deep sea effect. Some models maintain high-quality image capture in extreme lighting, including underwater and low-light settings—a feature referred to as the 'deep sea effect.' Enhanced processing algorithms enable applications in marine settings and low-visibility surveillance.

Facial recognition. Now widely adopted, facial recognition technology allows AI-powered cameras to identify individuals in seconds, even in crowds. Use cases range from secure building access to criminal investigations.

Predictive analytics. By analysing historical data, AI systems can forecast potential incidents. Unusual traffic patterns, suspicious movement, or crowd build-up can trigger proactive alerts—enabling early intervention.

Several well-known companies are at the forefront of integrating AI into their security solutions. Panasonic's AI cameras claim to detect abnormal human behaviour, enabling real-time alerts and more

effective crime prevention. Hikvision claims that their AI cameras come with features like automatic number plate recognition (ANPR), people counting, and heat maps, making them highly effective for retail, traffic monitoring, and smart city applications. Their approach includes both edge and cloud AI solutions, enabling flexibility depending on the user's needs. CP Plus is also claiming similar features for their AI cameras.

The future

Combining edge AI for real-time detection with server-based deep learning will further enhance surveillance. Cloud AI platforms like AWS Rekognition and Google Vision AI are enabling scalable, adaptable security solutions. Cameras are continuously improving as they learn from past incidents, boosting both accuracy and responsiveness.

Autonomous drones and robots with AI-enhanced vision will assist in patrolling and handling security threats. Simultaneously, compact edge AI cameras are making high-performance surveillance more accessible for small businesses and homes.

An exponential rise in the adoption of AI-powered cameras is anticipated. The potential of these technologies is only beginning to unfold.

Imagine retail signage adjusting in real time based on crowd mood, or smart city cameras detecting not just traffic offences but burnt-out lights or damaged road signs by reading and interpreting labels.

In short, the camera of tomorrow will not just see—it will understand. **EFY**

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